The infinite-dimensional Hilbert version of the relative spear theorem

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Source question

The source is Lakshmi Kanta Dey and Subhadip Pal, *On the geometry of G-norm*, arXiv:2603.20796. Question 1 on PDF page 13 asks:

> What is the infinite-dimensional version of Theorem 3.12?

Here Theorem 3.12 is the finite-dimensional Hilbert-space theorem asserting that a relative spear operator is a partial isometry. The source question is shown below.

> **Question 1.** *What is the infinite-dimensional version of Theorem 3.12?*
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> **Question 2.** *Can Theorem 3.12 and Theorem 3.13 be extended to the Banach space setting?*
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> **Question 3.** *If the previous answer is negative then find the operators G for which the conclusion of Theorem 3.12 continues to hold in the framework of general Banach spaces.*

Definitions

Let H_1, H_2 be Hilbert spaces over $\mathbb{K} = \mathbb{R}$ or $\mathbb{C}$, and use the inner product convention linear in the first variable. Let $G \in L(H_1, H_2)$ with $\|G\| = 1$. For $T \in L(H_1, H_2)$, set

$$\|T\|_G = \inf_{\delta>0} \sup\{\|Tx\| : x \in S_{H_1},\ \|Gx\| > 1 - \delta\}.$$

The G-numerical radius is

$$\nu_G(T) = \sup\{|\lambda| : \lambda \in V_G(T)\},$$

where $V_G(T)$ consists of the limits of $\langle Tx_i, y_i\rangle$ along nets with $x_i \in S_{H_1}$, $y_i \in S_{H_2}$ and $\Re\langle Gx_i, y_i\rangle \to 1$. We call G *relative spear* if

$$\nu_G(T) = \|T\|_G \qquad (T \in L(H_1, H_2)).$$

Theorem

Theorem 1. *Let H_1, H_2 be Hilbert spaces and let $G \in L(H_1, H_2)$ have norm one. Then G is relative spear if and only if H_2 is one-dimensional and G is a rank-one partial isometry onto H_2.*

Consequently, if $\|\cdot\|_G$ is required to be a genuine norm on $L(H_1, H_2)$, then G is relative spear if and only if H_1 and H_2 are both one-dimensional and G is a surjective isometry.

Proof intuition

The finite-dimensional theorem detects singular values in $(0,1)$ by mapping a top singular vector into a lower singular direction. In infinite dimensions the same obstruction has two forms. If the top spectral value of G^*G is isolated, the finite-dimensional argument works with the spectral projection at 1. If spectral values below 1 accumulate at 1, one uses two alternating families of near-top spectral slices and builds a nilpotent shift between them. That operator still has G-norm one because its domain slices approach the top of the spectrum, but its G-numerical radius is bounded by $1/2$.

Once G^*G is forced to be a projection, G is a partial isometry. A relative spear partial isometry cannot have missing final directions, because a test operator can point into the orthogonal complement of the final space and be invisible to ν_G. It also cannot have a final space of dimension at least two, because a two-dimensional nilpotent block has numerical radius $1/2$. Thus the only Hilbert relative spear case is one-dimensional in the codomain.

Proof

We first record a Hilbert-space simplification of V_G.

Lemma 1. *If $\Re\langle Gx_i, y_i\rangle \to 1$ with $x_i \in S_{H_1}$ and $y_i \in S_{H_2}$, then $\|Gx_i\| \to 1$ and $\|y_i - Gx_i\| \to 0$. Hence every element of $V_G(T)$ is a limit of $\langle Tx_i, Gx_i\rangle$ along a net with $\|Gx_i\| \to 1$.*

Proof. Since $\|Gx_i\| \le 1$ and $\|y_i\| = 1$, the condition $\Re\langle Gx_i, y_i\rangle \to 1$ forces $\|Gx_i\| \to 1$. Moreover

$$\|y_i - Gx_i\|^2 = 1 + \|Gx_i\|^2 - 2\Re\langle Gx_i, y_i\rangle \to 0.$$

Thus $|\langle Tx_i, y_i\rangle - \langle Tx_i, Gx_i\rangle| \le \|T\|\,\|y_i - Gx_i\| \to 0.$ □

Let $A = G^*G$. Then A is a positive contraction on H_1 with $\|A\| = 1$. We show first that A must be a projection.

Assume, toward a contradiction, that the spectral projection $E_A((0,1))$ is nonzero.

Case 1: the top is isolated from the rest of the spectrum. Suppose that 1 is isolated from $\sigma(A) \setminus \{1\}$. Then $M = E_A(\{1\})H_1$ is nonzero, and there is $\gamma < 1$ such that $\|A^{1/2}x\| \le \gamma\|x\|$ on $M^\perp$. Choose a unit $u \in M$ and a unit vector $v \in E_A((0,1))H_1$. Put $e = Gv/\|Gv\|$ and define

$$Tx = \langle x, u\rangle e.$$

Since $\|Gu\| = 1$ and $\|Tu\| = 1$, we have $\|T\|_G = 1$. If $\|Gx_i\| \to 1$, then writing $x_i = P_Mx_i + (I - P_M)x_i$ gives $\|(I-P_M)x_i\| \to 0$. Also $Gv \perp GM$, so

$$\langle Tx_i, Gx_i\rangle = \langle x_i, u\rangle\langle e, G(I-P_M)x_i\rangle \to 0.$$

By Lemma 1, $\nu_G(T) = 0$, contradicting relative spear.

Case 2: spectral values below 1 accumulate at 1. Choose pairwise disjoint Borel sets $I_n, J_n \subset (0,1)$ with nonzero spectral projections and with $\inf I_n, \inf J_n \to 1$. Pick unit vectors $u_n \in E_A(I_n)H_1$ and $v_n \in E_A(J_n)H_1$. The families (u_n) and (v_n) are mutually orthonormal, and the vectors $e_n = Gv_n/\|Gv_n\|$ are orthonormal in H_2. Define

$$Tx = \sum_{n=1}^{\infty} \langle x, u_n\rangle e_n.$$

This is a contraction. Since $\|Gu_n\| \to 1$ and $\|Tu_n\| = 1$, it follows that $\|T\|_G = 1$.

For every $x \in S_{H_1}$,

$$\langle Tx, Gx\rangle = \sum_n \langle x, u_n\rangle \langle e_n, Gx\rangle.$$

The coefficient vector $(\langle e_n, Gx\rangle)_n$ depends only on the projection of x onto $\overline{\operatorname{span}}\{v_n : n \geq 1\}$: indeed $\langle e_n, Gx\rangle = \langle Av_n, x\rangle/\|Gv_n\|$, and the mutually orthogonal vectors $Av_n/\|Gv_n\|$ have norm at most one because $0 \leq A \leq I$. Hence this coefficient vector has norm at most $\|P_V x\|$, and

$$|\langle Tx, Gx\rangle| \leq \|P_U x\| \, \|P_V x\| \leq \frac{1}{2},$$

where P_U and P_V are the orthogonal projections onto the closed spans of (u_n) and (v_n). Lemma 1 gives $\nu_G(T) \leq 1/2 < 1 = \|T\|_G$, again a contradiction.

Both cases are impossible; hence $E_A((0,1)) = 0$. Therefore $A = G^*G$ is an orthogonal projection, and G is a partial isometry. Let $M = \operatorname{ran} A$ be the initial space and $N = GM$ be the final space.

If $N \neq H_2$, choose a unit $u \in M$ and a unit $z \in N^\perp$. The operator $Tx = \langle x, u\rangle z$ satisfies $\|T\|_G = 1$ because $\|Gu\| = 1$. But $Gx \in N$ for all x, so $\langle Tx, Gx\rangle = 0$ for all x. Thus $\nu_G(T) = 0$, contradiction. Consequently $N = H_2$.

If $\dim H_2 \geq 2$, then $\dim M \geq 2$. Choose orthonormal $u_1, u_2 \in M$ and let S be the rank-one nilpotent on M given by $Su_1 = u_2$ and $S = 0$ on $u_1^\perp \cap M$. Define $T = GSP_M$, where P_M is the projection onto M. Then $\|T\|_G = 1$. For all $x \in H_1$,

$$\langle Tx, Gx\rangle = \langle SP_M x, P_M x\rangle.$$

The numerical radius of this nilpotent block is $1/2$, so $|\langle Tx, Gx\rangle| \leq 1/2$ for $x \in S_{H_1}$. Lemma 1 yields $\nu_G(T) \leq 1/2$, contradiction. Therefore $\dim H_2 = 1$.

Since $N = H_2$ and G is a partial isometry, G is a rank-one partial isometry onto the one-dimensional codomain.

Conversely, suppose $H_2 = \mathbb{K}v$ and

$$Gx = \langle x, u\rangle v$$

for unit vectors $u \in H_1$ and $v \in H_2$. Let $T \in L(H_1, H_2)$, so $Tx = f(x)v$ for some bounded functional f on H_1. If $x = \alpha u + h$ with $h \perp u$ and $|\alpha| > 1 - \delta$, then $\|h\| \leq (2\delta)^{1/2}$. It follows that

$$\|T\|_G = \lim_{\delta \downarrow 0} \sup_{\substack{x \in S_{H_1} \\ |\langle x, u\rangle| > 1-\delta}} |f(x)| = |f(u)|.$$

On the other hand, if $\Re\langle Gx_i, y_i\rangle \to 1$, then $y_i = \beta_i v$ with $|\beta_i| = 1$, $\langle x_i, u\rangle\overline{\beta_i} \to 1$, and $x_i - \langle x_i, u\rangle u \to 0$. Hence

$$\langle Tx_i, y_i\rangle = f(x_i)\overline{\beta_i} \to f(u).$$

Thus $\nu_G(T) = |f(u)| = \|T\|_G$ for every T, so G is relative spear.

Finally, if $\|\cdot\|_G$ is a genuine norm on $L(H_1, H_2)$ and $Gx = \langle x, u\rangle v$, then $u^\perp = \{0\}$; otherwise a nonzero operator supported on $u^\perp$ has G-seminorm zero. Hence H_1 is also one-dimensional. The converse in the one-dimensional case is immediate. $\square$

Answer to the Source Question

The infinite-dimensional Hilbert version of the source theorem is not merely "relative spear implies partial isometry." The exact classification is: relative spear Hilbert operators are precisely the rank-one partial isometries onto a one-dimensional codomain. With the source paper's standing convention that the G-norm is a norm on all of $L(H_1, H_2)$, the only Hilbert examples are one-dimensional surjective isometries.

Novelty Check and Limitations

Run-local indexes were searched for arXiv:2603.20796 and for the terms `G-norm`, `relative spear`, `partial isometry`, and Hilbert-space variants. No prior run packet was found. A targeted web search for the exact title and relative-spear/Hilbert/partial-isometry phrases did not locate an obvious later answer.

This packet answers Question 1. It does not settle the source paper's broader Banach-space Questions 2 and 3.

References

1. Lakshmi Kanta Dey and Subhadip Pal, *On the geometry of G-norm*, arXiv:2603.20796.